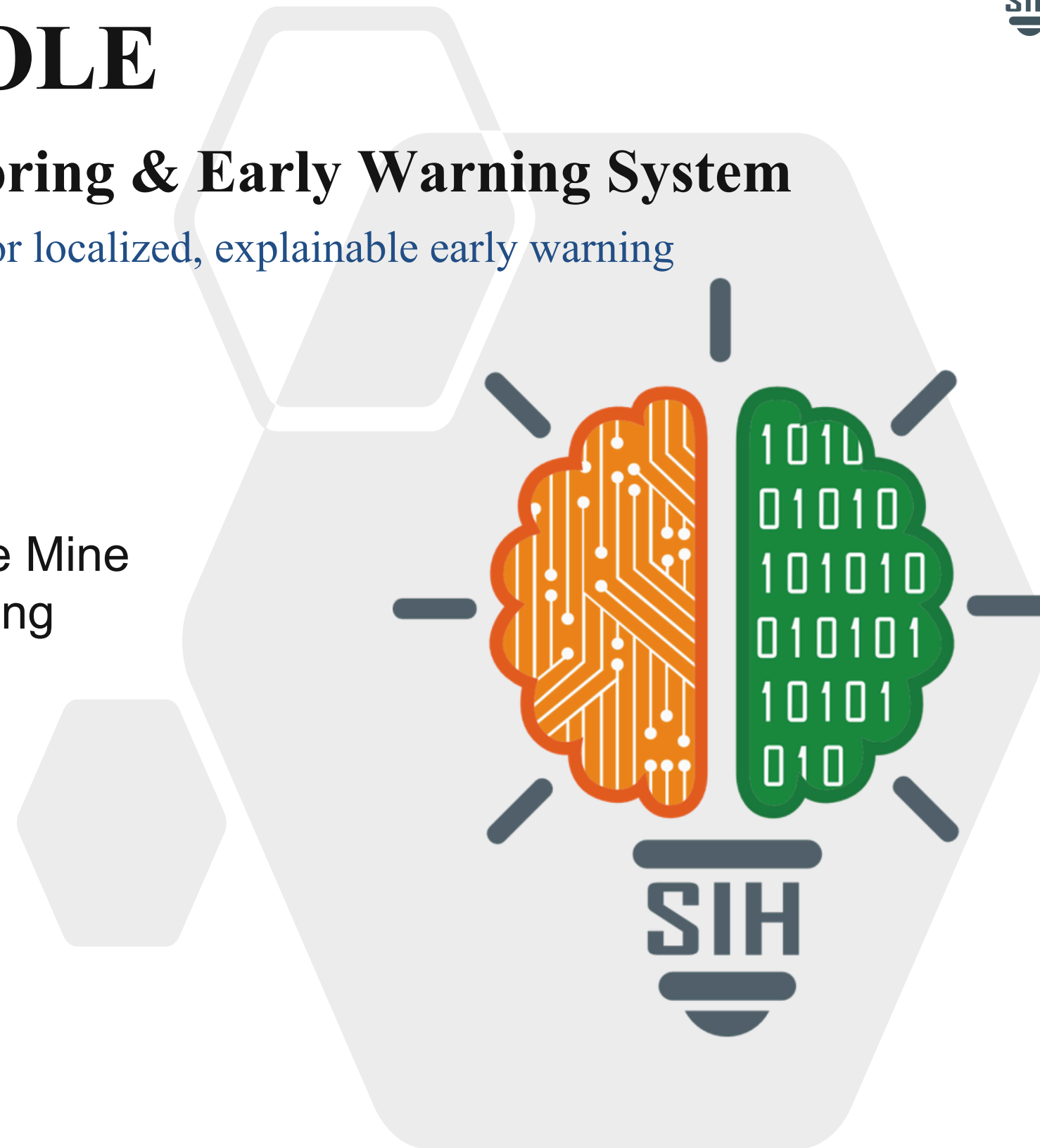


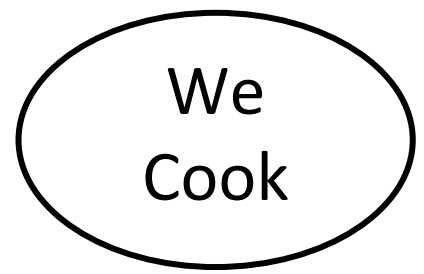
MOLE

Mine Subsidence Monitoring & Early Warning System

Distributed surface sensing for localized, explainable early warning

- **Problem Statement ID** – SIH26025
- **Problem Statement Title-**
Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India
- **Theme-** Smart Automation
- **PS Category-** Hardware
- **Team ID-** 157436
- **Team Name-** We Cook





IDEA TITLE

MOLE

Mine Observation & Live-alert Engine

Sense the ground. Predict movement. Warn early. Verify safely.



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HOW MOLE WORKS

CLOSED LOOP: every inspection improves the model



SOLUTION

- **6 Smart Sensor Nodes** → Real-time tilt, crack, vibration, gas & environmental monitoring
- **Edge AI + Multi-Node Fusion** → Anomaly detection & ground-movement prediction
- **Explainable Risk Score + GIS Heatmap** → Shows where & why risk is rising
- **Offline-First Alerts** → Reliable local intelligence without internet
- **Human-Approved Inspection Rover** → Camera + environmental sensing for verification
- **Closed-Loop Learning** → Verified rover data improves future predictions

NODE STACK

ESP32 platform across every node

- N1** MPU + slide pot + soil moisture
- N2** MPU + temperature
- N3** slide / linear + MQ-2
- N4** MPU + BME280 + LCD + buzzer
- N5** rain gauge + soil moisture
- N6** high-rate accel + piezo
- GW** Pi-class + LoRa concentrator
- RVR** 4WD ESP32 + camera + ultrasonic + gas/temp

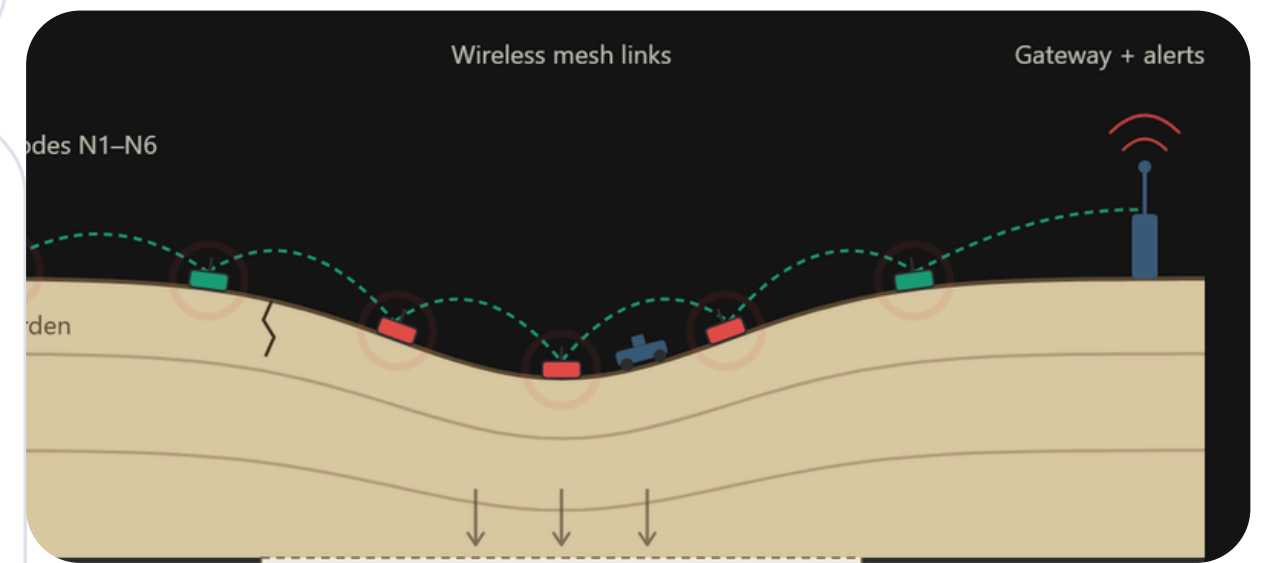
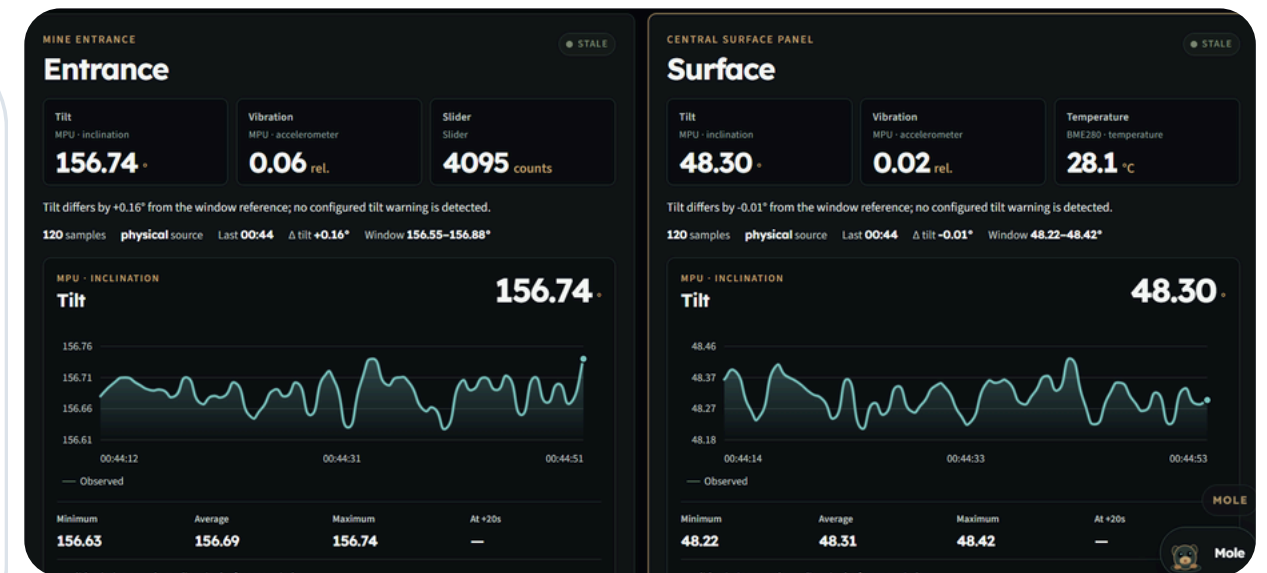
ML + TRUST

- 1 Rule engine = safety authority: WATCH / ALERT + persistence.
- 2 Isolation Forest learns each node; noisy one-off spikes are suppressed.
- 3 20 s / 20 min tilt-displacement forecasts with uncertainty.
- 4 Neighbour fusion + weather / drift health checks reduce false alarms.
- 5 Explainable alerts show node, channel, trigger and time window.

OFFLINE-FIRST

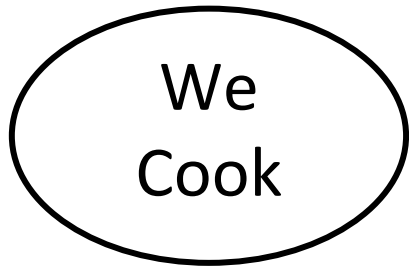
STALE ≠ SAFE

HUMAN-IN-THE-LOOP ROVER



PROOF LINKS
Demo Video — [paste link]
GitHub Repo — [paste link]
Live Dashboard — [paste link]

UNIQUE-SOLUTION EDGE AI+MULTI-NODE FUNCTION+ROVER
VERIFICATION+OFFLINE LLM IN ONE CLOSED-LOOP SAFETY SYSTEM



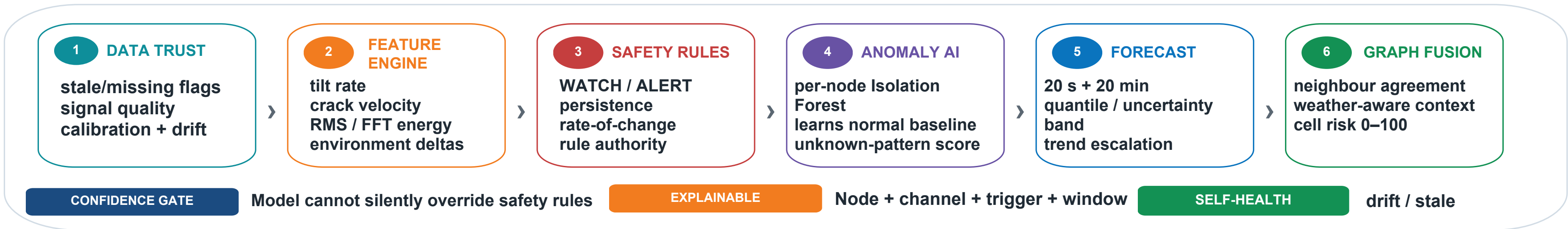
TECHNICAL APPROACH

Six-node sensing → edge AI → explainable risk → human-verified response



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EDGE AI / ML BRAIN



AI COPILOT

OFFLINE LLM + RAG — “Ask the mine”

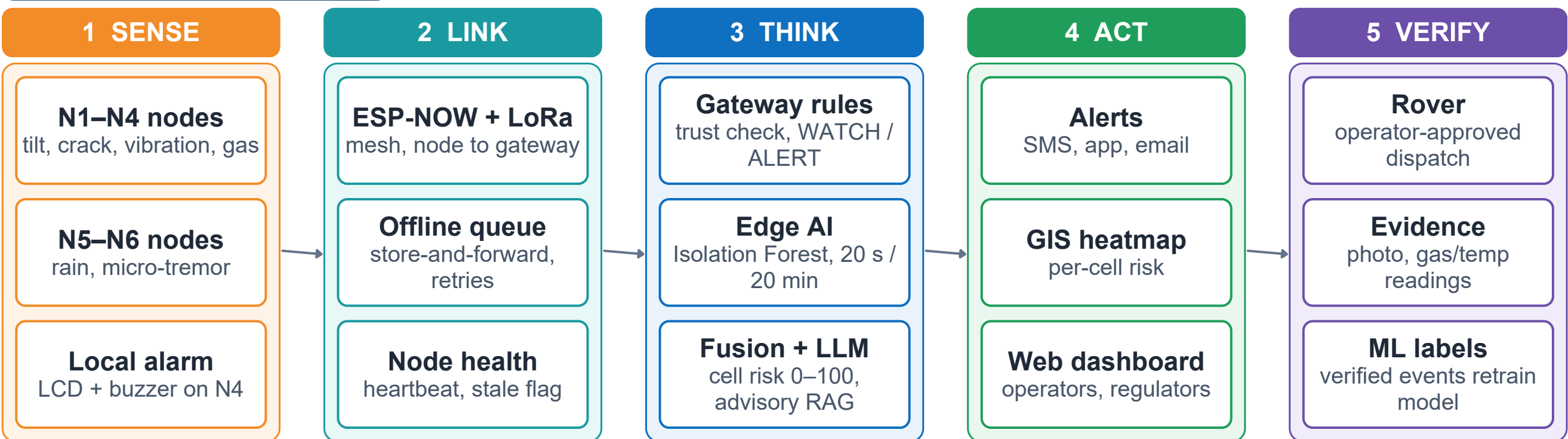
- Natural-language queries over live sensor data
- Retrieves SOPs + past incidents locally
- Explains why risk rose and drafts incident summaries
- Advisory only — never controls the alarm

INSPECTION ROVER

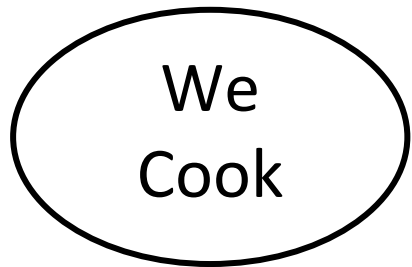
HUMAN-IN-THE-LOOP VERIFICATION

4WD ESP32 rover • camera • ultrasonic • gas + temperature
Operator dispatches only to the flagged cell → rover returns photo + local readings → verification is attached to the incident and stored as a future learning label.

SYSTEM ARCHITECTURE



CLOSED LOOP: rover-verified events return to the gateway as ML labels for model retraining



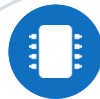
FEASIBILITY & VIABILITY

Modular by design • local-first for safety • scaled only after evidence gates



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TECHNICAL FEASIBILITY



COMMON NODE PLATFORM

ESP32-based modular nodes; N1–N4 are the current prototype, while N5–N6 extend rain + micro-tremor context in the pilot.



MULTI-PHYSICS SENSING

Tilt, vibration, crack-gap, wetness, gas, temperature, humidity, pressure and rainfall are fused instead of relying on one sensor.



OFFLINE-FIRST EDGE STACK

Gateway keeps rules, ML inference, local database, GIS and queued sync running even when internet is unavailable.



SAFE AI BOUNDARY

Rules remain the safety authority; anomaly/forecast models add confidence. The offline LLM is read-only and advisory.

BUILT-IN RESILIENCE



Heartbeat • stale-data flag • calibration / drift due



Prototype ESP-NOW → pilot LoRa + store-and-forward queue



Local LCD/buzzer continues even if cloud or laptop is down



Human-approved rover returns image + local readings as evidence

VIABILITY



MODULAR + REPLACEABLE

One failed node can be serviced without replacing the network.



SCALABLE BY ZONE

Same node/gateway architecture can add mine cells, IDs and GIS zones progressively.



LOWER HUMAN EXPOSURE

Rover verifies flagged ground only after operator approval; it is not a single point of failure.



AUDITABLE OPERATIONS

Timestamped telemetry, reason-coded alerts and incident evidence support review and maintenance.



LOW-COST MODULAR DESIGN

~₹1.5K per LoRa-enabled pilot node, depending on sensor configuration.

EVIDENCE-GATED SCALE-UP

No stage advances until data quality, communication, alarm behaviour and model performance are accepted.

1

TABLETOP PROTOTYPE



N1–N4 • ESP-NOW • dashboard
Scripted tilt / crack / vibration / environment events

2

CONTROLLED VALIDATION



Calibrate sensors • inject noise • packet-loss tests
Measure latency • false alarms • stale / drift handling

3

SURFACE PILOT



Add N5–N6 • LoRa • rugged enclosure • power plan
Mine-specific baseline + GIS zones + rover trials

4

FIELD HARDENING



Shadow-mode ML → reviewed influence on warnings
SOP integration • maintenance cycle • retraining gates

REAL RISKS → CONTROLS

DRIFT / NOISE

baseline + calibration flag
persistence + neighbour fusion

LINK / POWER

local alarm + queue + retries
watchdog + battery health

MODEL ERROR

rules first • confidence gate
shadow mode before influence

LOW LABELS

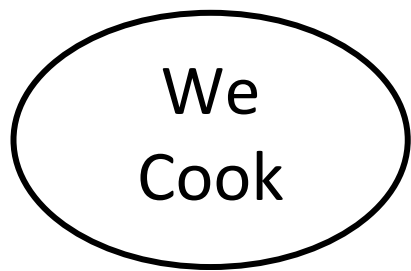
unsupervised anomaly detection
rover-verified events become labels

LLM HALLUCINATION

local RAG • source-grounded answer
read-only; never safety authority

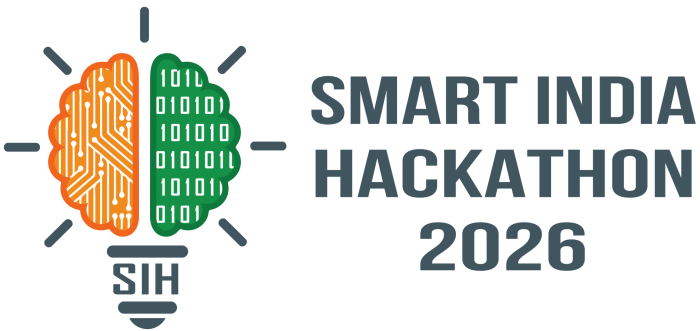
HARSH SITE

protected mounting + enclosure
replaceable node-level maintenance



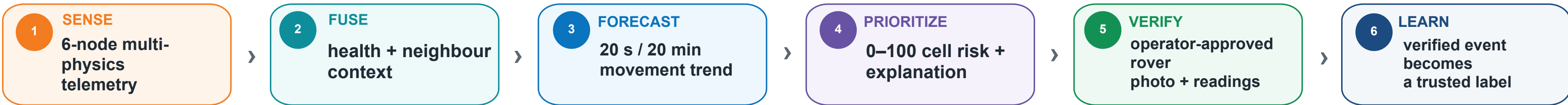
IMPACT & BENEFITS

From weak signals to verified action — earlier evidence, lower exposure, smarter mine decisions



IMPACT PATHWAY

MOLE closes the gap between sensing and a defensible field response.



WHO BENEFITS — HOW

SAFETY

Rover-first verification

Flagged zones can be checked with camera + gas/temperature context before sending people onto suspect ground. Local buzzer/LCD still works offline.

OPERATIONS

From periodic checks to continuous evidence

Live node health, stale-data flags and GIS risk ranking help teams focus inspection effort on the cells showing persistent, corroborated change.

ECONOMIC

Spend effort where evidence is strongest

Commodity modular nodes scale zone-by-zone; individual nodes are replaceable, while expensive survey/geotechnical work can be prioritized to flagged areas. low-cost nodes allow coverage to expand zone-by-zone instead of using expensive centralized instruments.

LAND + COMMUNITY

Protect what sits above the mine panel

GIS-linked risk cells make changes easier to relate to roads, rail, farms and settlements, while rainfall/soil context supports targeted follow-up.

PILOT SUCCESS

Measured — not claimed

Judge the system on engineering outcomes:

ALERT LATENCY

sensor → local warning

FALSE ALARMS

precision / event review

FORECAST ERROR

tilt / displacement MAE

NETWORK HEALTH

uptime + packet delivery

ROVER RESPONSE

alert → verified evidence

EXPLAINABILITY

alerts with clear cause

WHY MOLE

Not just DETECT → MOLE PREDICTS, EXPLAINS, VERIFIES and LEARNS — with local safety continuing even when the internet does not.

RESEARCH AND REFERENCES

Evidence → engineering decision → prototype proof

A. EVIDENCE THAT SHAPED MOLE

DOMAIN

[SIH26025 + CSIR-CIMFR](#)
Problem scope + Indian subsidence research → continuous deformation sensing, GIS risk zones and staged field validation.

SENSORS

[TDK MPU-6050 + Bosch BME280](#)
Datasheets anchor the node stack: 6-axis motion for tilt/vibration plus temperature, humidity and pressure context.

AI / ML

[Isolation Forest + feature engineering](#)
Unsupervised anomaly scoring complements rules; tilt-rate, displacement persistence, vibration RMS/FFT and neighbour context form the model features.

[Also referenced: CIMFR review of subsidence-monitoring techniques.](#)

SAFETY

[DGMS safety context](#)
Regulatory safety focus → human-authorized decisions, traceable alerts and rover verification instead of autonomous safety actions.

NETWORK

[ESP32 / ESP-NOW + LoRaWAN](#)
Prototype uses direct ESP32 communication; pilot migrates to a low-power long-range field network with store-and-forward resilience.

OFFLINE AI

[Local LLM inference](#)
Offline LLM/RAG explains alerts, retrieves SOPs and drafts incident summaries locally; it stays advisory and never overrides the safety engine.

[Pilot network reference: LoRa Alliance LoRaWAN specification.](#)

TRACEABILITY

SOURCE → REQUIREMENT → DESIGN CHOICE → TEST → FIELD VALIDATION
Every “advanced” feature is tied to a reference, a measurable signal and a validation step.

B. LINKS & VISUAL PROOF

- ▶ Demo Video — [paste link here](#)
- ▶ GitHub Repo — [paste link here](#)
- ▶ Live Dashboard — [paste link here](#)

Website / Dashboard
Screenshot

Prototype – Cave Model
(Full View)

Node Wiring /
Assembly Close-up

Working Demo /
Team Photo